

Iris Classification Machine Learning Project

Documentation

1. Project Overview

This project demonstrates a complete machine learning workflow for a multi-class classification task using the classic **Iris dataset**. The primary objective is to accurately classify iris flowers into their respective species based on physical measurements of their sepals and petals.

2. Dataset Description

- **Source File:** `iris.csv`
- **Total Instances:** 150 samples
- **Features:**
 - `sepal-length` (cm)
 - `sepal-width` (cm)
 - `petal-length` (cm)
 - `petal-width` (cm)
- **Target Classes** (50 samples per class):
 - `Iris-setosa`
 - `Iris-versicolor`
 - `Iris-virginica`

3. Methodology & Workflow

A. Exploratory Data Analysis (EDA)

Before training models, the dataset was explored to understand its structure and distribution:

- **Shape & Summary Statistics:** Inspected dataset dimensions (`150, 5`) and calculated summary metrics (mean, standard deviation, min/max, percentiles) for each feature.
- **Class Distribution:** Confirmed a balanced dataset with an equal distribution of 50 samples across all three target classes.
- **Visualizations:** Generated box-and-whisker plots, histograms, and a scatter plot matrix to observe feature correlations and distributions across different classes.

B. Data Preparation

- **Train-Test Split:** The dataset was split into training and validation sets using an 80/20 ratio (`test_size=0.20`, with a fixed random seed of `7` for reproducibility).

- **Feature/Target Separation:** Features (X) comprised the first four columns, while the target class (Y) comprised the final column.

C. Algorithm Spot-Checking

Six diverse classification algorithms were spot-checked and evaluated using 10-fold cross-validation with accuracy scoring:

1. **Logistic Regression (LR)**
2. **Linear Discriminant Analysis (LDA)**
3. **K-Nearest Neighbors (KNN)**
4. **Classification and Decision Tree (CART)**
5. **Gaussian Naive Bayes (NB)**
6. **Support Vector Machine (SVM)**

D. Model Training & Evaluation

- **Selected Model:** K-Nearest Neighbors (KNN)
- **Validation Performance:** The trained KNN model was evaluated on the unseen validation subset, achieving an overall accuracy of **90%**.
- **Detailed Metrics:**
 - **Precision & Recall:** High performance across all classes, with *Iris-setosa* achieving perfect precision and recall (1.00), and strong metrics for *Iris-versicolor* and *Iris-virginica*.
 - **Confusion Matrix:** Demonstrated minimal misclassifications, correctly identifying 7 out of 7 setosas, 11 out of 12 versicolors, and 9 out of 11 virginicas.

4. Conclusion

The machine learning pipeline successfully classified the Iris flower dataset with high accuracy. The KNN classifier proved to be an effective model for this classification problem, supported by clean data distributions and robust cross-validation scores.